Neural Correlates of Short-Video Application Addiction and Adolescents' Executive Functions

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Abstract: Considering the widespread influence of short-video applications (SVAs) on adolescent learners, this study aimed to examine whether and how problematic SVA use predicts their executive functions from behavioral and neural perspectives. With a sample of Chinese adolescents (M_{age}=12.30, N=84), behavioral results show that SVA addiction predicted lower working memory but not cognitive control performance. Their brain activation pattern during SVA use was also collected and linear regression analysis generated a neural indicator of SVA addiction on delta band. This component is proved to be a significant negative predictor of adolescents' working memory and cognitive control performance. Both behavioral and neural evidence provide complementary perspectives on the relationship between SVA addiction and adolescents' executive functions. The findings suggest that though SVAs have played increasingly important role in adolescents' informal learning experiences, the negative association between SVA overuse and their cognitive functioning urges caution.

Introduction

Short video application (SVA), a newly-rising digital media giant that features real-time shooting and sharing of short videos on social platforms, has been growing dramatically worldwide in the past few years. Noticeably, adolescents are playing a substantial role among the huge quantities of users of SVAs. The interactivity and creativity of SVAs have brought profound changes to their learning experiences through facilitating and equalizing the way knowledge is produced and shared without explicit control or judgment (Dezuanni et al., 2022; UNESCO, 2022). SVAs have also provided learners with ample opportunities of connected learning beyond formal schooling (Ito et al., 2019), which may empower adolescents to become more independent and capable learners. In the meantime, skeptical views on the potential negative impact of SVA use on cognitive development are increasing (Lövdén et al., 2020; Sha & Dong, 2021) and even proved by a recent empirical study on adolescents (Xu et al., 2023). Moreover, adolescents' overindulgence in SVA use has also become prominent and aroused public concern over the influence of problematic SVA use on their executive functions (Lu et al., 2022; Marengo et al., 2022), which are significant predictors of their learning performance (Gathercole et al., 2004). However, there is still lack of solid evidence on this relationship and more importantly, on how this impact might take place during SVA use. Thus, this study aims to address the potential impacts of SVA addiction on adolescents' executive functions from both behavioral and neural perspectives. The results may help adolescents and educators to be better informed of the potential negative impact of SVA overuse and make reasonable decisions on the proper use of SVAs.

Literature review

Potential impact of SVA addiction on adolescents' executive functions

The ephemerality of short-video form, powerful algorithmic recommender and attractive audiovisual experiences have made SVA users more likely to indulge themselves and even become addicted to it (Huang et al., 2022; Xu et al., 2023). Also, the functional hooks facilitate uncontrolled and automatic use of SVAs (Wang & Scherr, 2021; Yang et al., 2021). Thus, repetitive exposure may contribute to SVA addiction, which is the major concern regarding adolescents' SVA use (Marengo et al., 2022).

The potential impact of SVA addiction on adolescents is primarily on their cognitive development, as the constant intrusion of mobile technology on cognition may revolutionize how they process information and regulate attention (Chen et al., 2022; Wilmer & Chein, 2016; Zhao, 2021), especially at the key stage for cognitive development during adolescence (Sawyer et al., 2018). Working memory capacity, one of the core executive functions (EFs), is regarded as susceptible to SVA addiction as habitually watching highly-personalized videos may dispose users towards passive consumption with less cognitive control or reflections taking place in working memory (Hasan et al., 2018; Zhao, 2021). Indeed, SVA users browse the diverse personalized content and react to audio-visual stimuli at the same time, which may boost the demand for information processing in working memory and induce cognitive enhancement (Colzato et al., 2013; Waris et al., 2019). However, fast-paced SVs full of visual edits and cuts may increase involuntary attention and arousal, which can cause information overload for users {Ma, 2022 #495}. This may fundamentally alter people's way of obtaining and storing information, as evidenced by the negative relationship between TikTok use disorder and short-term memory (Sha & Dong, 2021). For adolescent learners, working memory plays a crucial role of storing and manipulating information during learning tasks, which can strongly predict adolescent students' academic performance on various subjects (Best et al., 2011; Grimley & Banner, 2008).

Also, adolescents' cognitive control may be impaired through excessive SVA use, which refers to the top-down control required to select the goal-relevant task in the face of competing alternatives (Grange & Houghton, 2014). Previous studies on problematic media use and multitasking suggest that individuals with addictive behaviors tend to have difficulties in focusing attention on the task and resisting the temptation of various distractions (Chen et al., 2016; Kong et al., 2023; Mazhari, 2012). Additionally, neuroimaging studies show that the brain regions related to cognitive control are usually activated among individuals with behavioral addiction since more cognitive resources are required and employed to accomplish the tasks (Darnai et al., 2019). For SVAs, constant media multitasking and immediate gratification of entertainment-seeking may influence users' cognitive control of attention, namely selective or focused attention. This is proved by the positive relationship between attention deficits and SVA addiction (Chen et al., 2022). However, heavy media multitaskers among late adolescents were found to perform better in switching between tasks (Alzahabi & Becker, 2013), suggesting that vast experience in multitasking with the dynamic and complex media activities may enhance their capacity of rapidly shifting between tasks and guiding goal-directed behaviors. Cognitive control is particularly important for learners in the digital age, as it reflects students' use of strategies related to self-regulated learning in order to sustain motivation and academic goals (Kokoç, 2021).

The above review suggests that there still exists inconsistency in current literature as to whether adolescents' SVA addiction may negatively correlate with their executive functions. Moreover, most current studies on SVA addiction focus on the contributing factors (Zhang et al., 2019) while there is little evidence on its potential impact on cognition. Thus, this study examines whether and how SVA addiction may influence adolescents' working memory and cognitive control.

Neural evidence on the potential impact of SVA addiction on executive functions

In light of behavioral evidence on the potential change in executive functions brought by media use and the crucial role of brain plasticity in adolescents' cognitive development (Li & Li, 2010), neuroscientific studies are expected to discover the neural underpinnings of how media use exerts its impact.

Electroencephalogram (EEG) has been widely adopted to record the electrical activity of brain as a useful tool to find out the neural correlates of addictive disorders regarding media use, including problematic Internet use (D'Hondt et al., 2015) and Internet gaming disorder (IGD) (Hong et al., 2022). Meanwhile, EEG oscillatory responses to stimuli could be general electrophysiological markers for cognitive function and dysfunction. For instance, delta oscillatory responses are involved in cognitive processes including attention, perception, and decision-making, of which the decrease is found associated with Alzheimer's disease, schizophrenia and alcoholism (Güntekin & Başar, 2016). However, only a few studies examine the brain activity during media use, which can further reveal the potential mechanism of how media use impacts cognitive processing. Compared with the abundant evidence on addiction-related resting-state EEG features, there is lack of evidence on the brain activity during SVA use on both regular and problematic users, which may provide insights into how SVA use may exert its impact on adolescents' cognition. Thus, the current study intends to investigate the brain activation during SVA use and further explore its relationship with adolescents' executive functions.

Methodology

Participants

The participants of the study were 84 adolescents (M_{age}=12.30, S.D.=0.51, 43 males) recruited from a middle school in Beijing. Informed consent was obtained from the participants and their legal guardians before data collection. All the procedures involved in data collection was carried out during school days under the supervision of trained researchers, and the study protocol was approved by the IRB of the Department of Psychology, Tsinghua University.

Behavioral measures

SVA addiction scale is adapted from Internet Addiction Test (Pawlikowski et al., 2013; Young, 1998). The test is confirmed with high reliability (Cronbach'α=0.931) and good model fit (χ^2/df=3.115, CFI=0.974, RMSEA=0.054).

Working memory is assessed by adapted operation span task (Turner & Engle, 1989). For each trial, participants must solve a series of arithmetic equations while trying to remember a list of Chinese characters. Cognitive control was assessed by task switching paradigm, where the participants were asked to tell which number was larger even though the numbers were presented in different sizes (Logan, 2003).

EEG acquisition and preprocessing

One week after the behavioral measures, the participants were invited to the EEG recording session. The EEG recording system includes an 8-channel EEG cap (F3, F4, T3, T4, P3, P4, O1, O2) and a DC-coupled amplifier (sampling rate at 500 Hz).

During the recording session, each participant was seated on a comfortable chair in an isolated classroom and was asked to follow the instructions and limit body movements. After 1-minute resting state as baseline, the participant was asked to freely browse SVs according to user preference on TikTok website (www.douyin.com) in 3 minutes. EEG data was down-sampled to 250 Hz and subsequently detrended to remove the DC component. After bandpass filtered at 1-50 Hz, data was re-referenced to common average and segmented into 2-s epochs. After removing artifacts, the remaining epochs were averaged in delta (0.5–4 Hz), theta (5–8 Hz), alpha (9–12 Hz), beta (13–30Hz), and gamma (31-45Hz). Relative power was adopted in analyses to normalize data (Houston & Ceballos, 2013; Nishiyori et al., 2021).

Statistical analysis

Brain activation during SVA use is calculated by subtracting the baseline from EEG band power during short-video watching. For each band, multiple linear regression was performed to find the correlation between EEG activation and SVA addiction. The SVA addiction score was used as the dependent variable and the band power of 8 channels served as predictors. After identifying the EEG components correlated with SVA addiction, their relationship with the test scores regarding the executive functions was examined.

Results

Behavioral results

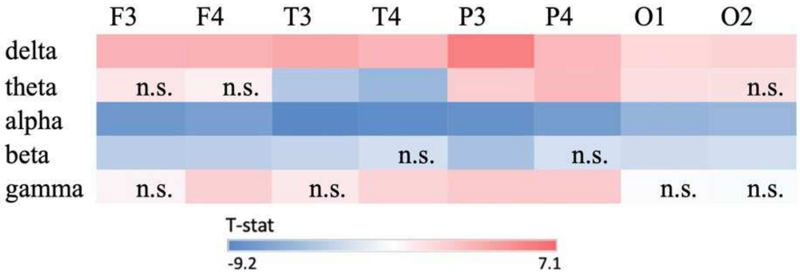
In terms of the relationship between cognitive test performance and SVA addiction, SVA addiction shows significant negative relationship with working memory (r= -0.340, p=0.003) but not with cognitive control (r= -0.234, p=0.080).

Neural results

The neural changes during short-video watching are displayed in Figure 1. Compared to baseline, adolescents showed whole-brain increase in delta activity and decrease in alpha and beta band. Theta activity was enhanced in parietal and left occipital regions while reduced in temporal regions. Gamma activity experienced increase in right frontal, temporal and parietal regions as well as in left parietal region.

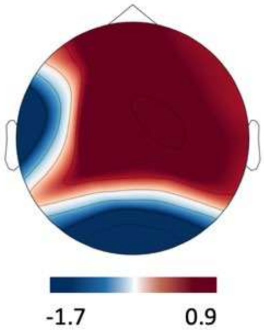
Figure 1

The Averaged Brain Aactivation During Short-video Watching Task



Note. The shades of color in this figure denotes the t-statistics of the mean difference in brain activities between task condition and baseline. Apart from the insignificant differences (shown as 'n.s.'), the other differences are statistically significant (p<0.05).

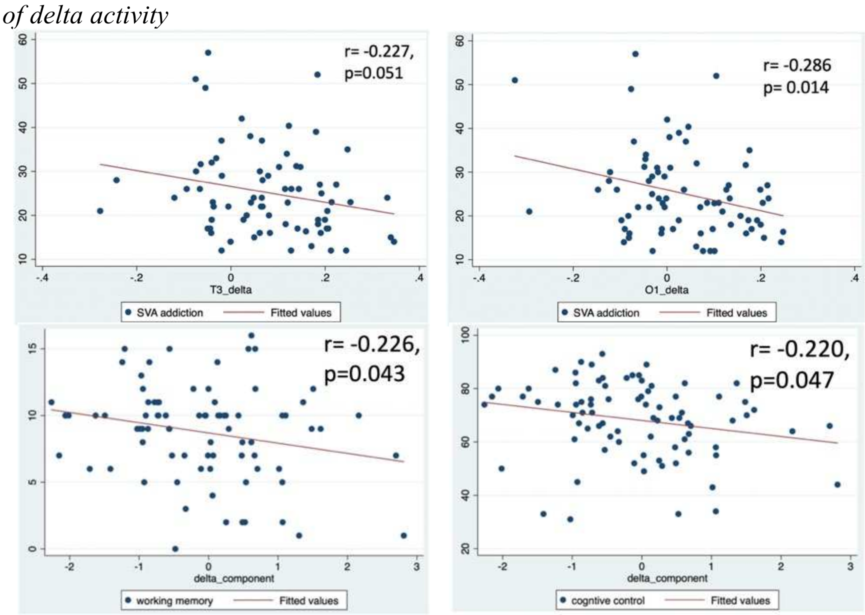
Figure 2
Topographic Display of Standardized Coefficients.



SVA addiction was only found significantly predicted by the change in delta band power during the video-watching task (F= 2.184, p= 0.040, adjusted R^2=0.115). The standardized coefficients were displayed in Figure 2. The regression analysis on other band power (i.e., theta, alpha, delta, gamma) did not generate significant results. Thus, the delta activation pattern was regarded as the neural indicator of adolescents' SVA addiction for subsequent analysis. According to this pattern, the changes in left temporal and occipital delta activity were found negative related to SVA addiction (see Figure 3).

Meanwhile, this neural indicator predicted lower working memory (r= -0.226, p=0.043) and cognitive control performance (r= -0.220, p=0.047) among participants (see Figure 3).

Figure 3
Relationship between Executive Function and the SVA-addiction-related Pattern of delta activity



Conclusion and discussion

Concerning the prevalence of SVAs in adolescents' learning experiences and the problematic use of SVAs among them who are experiencing crucial cognitive development, the present study addresses the negative association between adolescent users' SVA addiction and their executive functions from both behavioral and neural evidence. The results indicate that while affording digital literacy development and collaborative learning (Boffone, 2022; De Leyn et al., 2021), SVAs may also pose challenges to young learners' cognitive development.

The behavioral findings of this study show that SVA addiction can be a negative predictor of adolescent users' working memory, which is consistent with the previous findings on the negative relationship between adolescents' SVA user behaviors and their working memory capacity (Xu et al., 2023). Thus, addictive use of

SVA may be associated with impaired ability to store and process information simultaneously, as well as a comprehensive and strategic use of working memory resources (Murphy & Creux, 2021). However, problematic SVA use did not show significant negative relationship with cognitive control. Since the current study used tasking-switching paradigm to measure this executive function, the result suggests that tasking switching and cognitive flexibility may not be affected by SVA use.

From neurophysiological perspectives, this study discovers a neural indicator of adolescents' SVA addiction on delta activity during SVA use, which was found negatively correlated with worse working memory and cognitive control. Thus, delta oscillations during SVA use may reflect individual's tendency towards addiction, which generally reflect attentional engagement (Klimesch et al., 2007), automatic regulation and reward processing (Knyazev, 2012). The enhancement of delta activity in task-EEG is associated with increased attention to the stimuli, while the impairment of slow wave-related responses may indicate lower cognitive involvement and reward deficiency. Overall, the adolescents showed increase in delta power during short-video watching compared to baseline EEG, suggesting that short videos could evoke higher attentional engagement. However, the more addictive adolescents are to SVA use, their left occipital and temporal delta activities are less raised and even decrease. This coincides with the neural evidence on chronic alcohol or drug use, suggesting that decrease in delta power in response to the stimuli may be induced by sensitization (Alper, 1999). Thus, for those who are more prone to addiction, SVA use may demand less attentional engagement and induce higher sensitivity and impulsivity, especially in the processing of audio-visual information. Since many learning materials are displayed in the audio-visual form in the digital era, the excessive use of SVAs may the young learners to lack of attentional control and more superficial way of information processing (Young & De Abreu, 2017). Meanwhile, this SVA-addiction-related neural pattern is a negative predictor of their working memory and cognitive control, which is aligned with the previous finding that the impairment of occipital lobe with additional damage to the temporal regions was correlated with executive dysfunction across many domains (Park et al., 2011). This correlation also indicates that the increased sensitivity and impaired reward-processing during SVA use can predict worse executive functions among adolescents.

Furthermore, the discrepancy between behavioral and neural findings in the current studies proves that the two different measures can be different and complementary for the identification of the potential impact of SVA addiction on adolescents' cognitive functioning. Since traditional cognitive assessments merely generate the behavioral outcomes, the ongoing or small potential impact may not be easily traced. In this sense, this study provides new perspectives on investigating the relationship between adolescents' addictive media use and their cognitive development, and more evidence is required on this issue for further discussion.

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